

Steps of the Machine Learning

- Data Collection
- EDA
 - Data Cleaning:
 - Data Analysis
 - Pre-processing
- Split the Data in to Training and Testing
- Transform
 - Encoding
 - Scaling
- Model Building
 - Model training
 - Model Evaluation
 - Prediction Pipeline

Model Building

Training

Prediction

Training Phase

- a) Choose model based on problem
- b) Hyper parameter tuning
- c) Model evaluation
- d) Select the best model
- e) Save model as pkl file

Prediction Phase

- a) Create the API to predict the new data
- b) Handover the API to user (aws, or any other cloud)
- c) API ready for prediction

x						y
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CustID	Age	Monthly Spend	Tenure Months	Support Calls	Payment Method	Return
1001	56	79.7	22	3	Bank Transfer	Yes
1002	46	63.38	44	0	Bank Transfer	No
1003	32	38.57	25	2	Bank Transfer	No
1004	60	104.62	27	4	Credit Card	No
1005	25	27.33	42	2	Bank Transfer	Yes
1006	38	113.86	28	4	Credit Card	No
1007	56	142.01	16	0	Bank Transfer	Yes
1008	36	20.1	15	1	Electronic Check	No
1009	40	148.99	47	3	Electronic Check	No
1010	28	100.27	44	0	Credit Card	Yes

- EDA

- 1) Data Cleaning:
- 2) Data Analysis
- 3) Pre-processing

X= Independent Variable

Y= Dependent Variable

x						y
CustID	Age	Monthly Spend	Tenure Months	Support Calls	Payment Method	Return
1001	56	79.7	22	3	Bank Transfer	Yes
1002	46	63.38	44	0	Bank Transfer	No
1003	32	38.57	25	2	Bank Transfer	No
1004	60	104.62	27	4	Credit Card	No
1005	25	27.33	42	2	Bank Transfer	Yes
1006	38	113.86	28	4	Credit Card	No
1007	56	142.01	16	0	Bank Transfer	Yes
1008	36	20.1	15	1	Electronic Check	No
1009	40	148.99	47	3	Electronic Check	No
1010	28	100.27	44	0	Credit Card	Yes

X= Independent Variable
Y= Dependent Variable

X_train

CustID	Age	Monthly Spend	Tenure Months	Support Calls	Payment Method
1001	56	79.7	22	3	Bank Transfer
1002	46	63.38	44	0	Bank Transfer
1003	32	38.57	25	2	Bank Transfer
1004	60	104.62	27	4	Credit Card
1005	25	27.33	42	2	Bank Transfer
1006	38	113.86	28	4	Credit Card
1007	56	142.01	16	0	Bank Transfer

y_train_true

Return
Yes
No
No
No
Yes
No
Yes

- Split the Data in to Training and Testing
 - Training 70%
 - Testing 30%

X_test

CustID	Age	Monthly Spend	Tenure Months	Support Calls	Payment Method
1008	36	20.1	15	1	Electronic Check
1009	40	148.99	47	3	Electronic Check
1010	28	100.27	44	0	Credit Card

y_test_true

Retrun
No
No
Yes

Data After Encoding

x_train

Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
56	79.7	22	3	1	0	0
46	63.38	44	0	1	0	0
32	38.57	25	2	1	0	0
60	104.62	27	4	0	1	0
25	27.33	42	2	1	0	0
38	113.86	28	4	0	1	0
56	142.01	16	0	1	0	0

x_test						
36	20.1	15	1	0	0	1
40	148.99	47	3	0	0	1
28	100.27	44	0	0	1	0

Scaling

X_Train						
Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
0.91	-0.04	-0.75	0.55	1	0	0
0.1	-0.47	1.56	-1.38	1	0	0
-1.02	-1.11	-0.44	-0.09	1	0	0
1.23	0.61	-0.23	1.2	0	1	0
-1.58	-1.4	1.35	-0.09	1	0	0
-0.54	0.85	-0.12	1.2	0	1	0
0.91	1.58	-1.38	-1.38	1	0	0

X_test						
Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
0.27	-1.31	-1.41	-0.27	0	0	1
1.07	1.11	0.81	1.34	0	0	1
-1.34	0.2	0.6	-1.07	0	1	0

x_train

Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
0.91	-0.04	-0.75	0.55	1	0	0
0.1	-0.47	1.56	-1.38	1	0	0
-1.02	-1.11	-0.44	-0.09	1	0	0
1.23	0.61	-0.23	1.2	0	1	0
-1.58	-1.4	1.35	-0.09	1	0	0
-0.54	0.85	-0.12	1.2	0	1	0
0.91	1.58	-1.38	-1.38	1	0	0

Model
Training

Model.pkl

x_train

Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
0.91	-0.04	-0.75	0.55	1	0	0
0.1	-0.47	1.56	-1.38	1	0	0
-1.02	-1.11	-0.44	-0.09	1	0	0
1.23	0.61	-0.23	1.2	0	1	0
-1.58	-1.4	1.35	-0.09	1	0	0
-0.54	0.85	-0.12	1.2	0	1	0
0.91	1.58	-1.38	-1.38	1	0	0



Model.pkl



y_train_pred

Return
1
1
1
0
1
0
1

training_x

Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
0.91	-0.04	-0.75	0.55	1	0	0
0.1	-0.47	1.56	-1.38	1	0	0
-1.02	-1.11	-0.44	-0.09	1	0	0
1.23	0.61	-0.23	1.2	0	1	0
-1.58	-1.4	1.35	-0.09	1	0	0
-0.54	0.85	-0.12	1.2	0	1	0
0.91	1.58	-1.38	-1.38	1	0	0

y train_true

Return
1
0
0
0
1
0
1

y_train_pred

Return
1
1
1
0
1
0
1

x_test

Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
0.27	-1.31	-1.41	-0.27	0	0	1
1.07	1.11	0.81	1.34	0	0	1
-1.34	0.2	0.6	-1.07	0	1	0

y_test_true

Return
0
0
1



Model.pkl



y_test_pred

Return
1
0
1

x_train

Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
0.91	-0.04	-0.75	0.55	1	0	0
0.1	-0.47	1.56	-1.38	1	0	0
-1.02	-1.11	-0.44	-0.09	1	0	0
1.23	0.61	-0.23	1.2	0	1	0
-1.58	-1.4	1.35	-0.09	1	0	0
-0.54	0.85	-0.12	1.2	0	1	0
0.91	1.58	-1.38	-1.38	1	0	0

Y_train_true

Return
1
0
0
0
1
0
1

Y_train_pred

Return
1
1
1
0
1
0
1

x_test

Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
0.27	-1.31	-1.41	-0.27	0	0	1
1.07	1.11	0.81	1.34	0	0	1
-1.34	0.2	0.6	-1.07	0	1	0

y_test_true

Return
0
0
1

Y_test_pred

Return
1
0
1

Evaluation Phase

Training Data Evaluation

Testing Data Evaluation

Training
Data Evaluation

Actual_
training_y

Return
1
0
0
0
1
0
1

Training_
predicted_y

Return
1
1
1
0
1
0
1

Testing
Data Evaluation

Actual_
testing_y

Return
0
0
1

predicted_
testing_y

Return
1
0
1

Data Collection

EDA

- 1) Data Cleaning:
 - 2) Data Analysis
 - 3) Advance Pre-processing
 - Handel Imbalance Data
 - Feature Selection
 - Dimension Reduction
 - 4) Encoding
 - 5) Split the data Training and Testing
 - Training Data 80%,
 - Testing Data 20%
- Scaling
 - Model Building
 - Model training
 - Model Evaluation
 - Prediction Pipeline